

Regression Exam Assignment

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1 Introduction

This report investigates the relationship between cardiovascular disease (CVD) and its major risk factors using data from the *Framingham Heart Study*, a pioneering longitudinal study initiated in 1948 to uncover the causes of heart disease. The dataset contains information on participants observed across up to three examinations and followed for 24 years. It includes demographic, lifestyle, and clinical variables such as age, sex, blood pressure, cholesterol, smoking, diabetes, and body mass index, along with indicators of cardiovascular events and survival times.

The primary aim of this report is to model and interpret how these factors influence both the **occurrence** (CVD) and **timing** (TIMECVD) of cardiovascular events. The analysis proceeds in three stages: an exploratory data analysis to understand data structure and relationships, binary regression modeling to estimate disease risk, and survival analysis to account for event timing and censoring.

2 Exploratory data analysis

To begin our analysis, we first explore and understand the structure and quality of the dataset before modeling. The exploratory data analysis aims to summarize the main characteristics of the data and their distributions, identify missing values or inconsistencies, and inspect relationships among key variables.

The dataset includes 11,627 observations across 39 variables, with each of the 4,434 participants appearing in up to three examination periods. Our focus in this stage is not on modeling outcomes yet, but on developing a clear overview of the available predictors, understanding their ranges and types, and assessing how complete and reliable the data are. This foundation ensures that the regression and survival analyses that follow are based on well-understood and well-prepared data.

2.1 Missing Values

We begin by examining the extent and structure of missing values across all 39 variables in the Framingham dataset. Missingness was first summarized using the command `colMeans(is.na(df)) * 100`, which calculates the percentage of missing entries for each variable. Overall, the dataset is highly complete: most variables show less than 5% missingness. However, a few variables stand out. The lipid measures HDLC and LDLC have approximately 73.97% missing values, while GLUCOSE and BPMEDS exhibit moderate levels of missingness at 12.38% and 5.10%, respectively. Smaller gaps are seen for TOTCHOL (3.52%) and educ (2.54%), while key demographic, clinical, and outcome variables such as AGE, SEX, SYSBP, DIABP, and CVD are fully observed.

To better understand the pattern of missingness, we compared the availability of key variables across examination periods. Using the command `table(df$PERIOD[!is.na(df$HDLC)])`, we found that HDLC and LDLC measurements were recorded exclusively in Period 3, being completely unmeasured (100% missing) in Periods 1 and 2. This structural missingness is expected since lipid profiling was introduced only in later examination cycles. For other variables such as GLUCOSE, missingness increases gradually across periods, as shown by `round(100 * tapply(is.na(df$GLUCOSE), df$PERIOD, mean), 2)`, which gives approximately 8.95%, 12.24%, and 17.22% missingness in Periods 1, 2, and 3, respectively. This pattern likely reflects participant drop-out over time rather than measurement error, which is common in long-term cohort studies.

Since the overall proportion of missing values is relatively low and shows no strong dependency on other observed variables, we assume that the non-structurally missing data is missing completely at random to a reasonable degree. Therefore, we proceed with a **complete-case analysis** at baseline (PERIOD = 1), excluding individuals with any missing predictor values. This decision simplifies the analysis and avoids introducing unnecessary assumptions about the missingness mechanism. The complete-case dataset still retains the majority of participants and provides a consistent basis for comparison across predictors.

We also inspected missingness patterns visually using bar plots and missingness maps. Missing values appeared concentrated in HDLC and LDLC (structural missingness) and were otherwise scattered randomly across individuals, with no clear association with age or sex. This supports the assumption that the remaining data are approximately missing at random.

Observations with missing HDLC and LDLC deleted only if they are missing during period 3.

With missingness addressed and incomplete observations removed, we now explore the marginal distributions of the main predictors.

2.2 Marginal Distributions

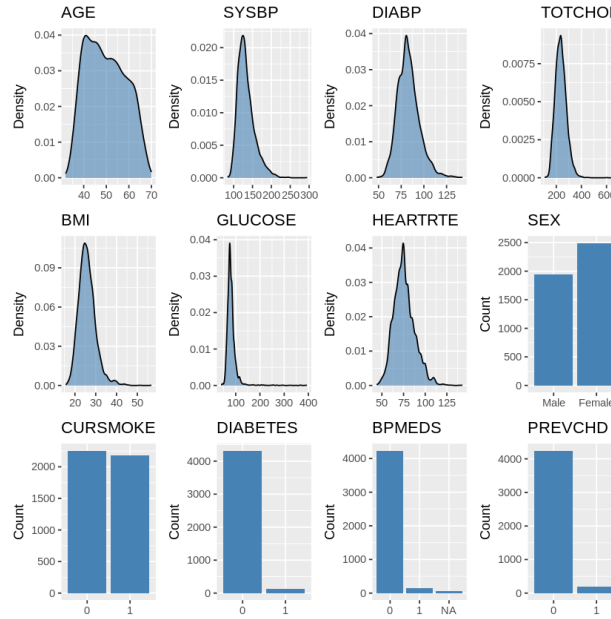
To characterize the study population, we examined the marginal distributions of the main predictors at baseline (PERIOD = 1). Restricting the analysis to baseline ensures that each participant is represented only once and that the distributions describe the population at study entry, before any follow-up changes in risk factors or disease status occurred. Furthermore the predictive model we will develop uses data from Period 1 only.

The combined plot below shows density curves for continuous variables alongside bar charts for categorical variables. Most continuous predictors are approximately symmetric or mildly right-skewed. AGE, SYSBP, and DIABP display bell-shaped distributions centered around 50–60 years and 120–140 mmHg, which reflect a predominantly middle-aged population with blood pressure in the upper normal range. In contrast, GLUCOSE, CIGPDAY, and TOTCHOL show strong right tails, indicating that a minority of participants have particularly high glucose levels, heavy smoking habits, or elevated cholesterol, factors that may contribute to increased CVD risk. These observations align with the original Framingham findings that such traits cluster among individuals at higher cardiovascular risk.

For categorical variables, we observe that slightly more than half of the sample are women, and that most participants are non-diabetic and not on blood pressure medication. Around 40% are current smokers and only a small fraction had prevalent coronary heart disease (PREVCHD) at baseline, confirming that the cohort largely consisted of initially healthy adults. This provides an appropriate foundation for studying incident cardiovascular disease over time.

Although several predictors exhibit moderate skewness, we decided not to transform them. The goal of this step is descriptive understanding rather than normalization, and keeping the original scales preserves interpretability of model coefficients in later regression analyses.

```
p1 <- subset(df, PERIOD == 1)
continuous <- c("AGE", "SYSBP", "DIABP", "TOTCHOL", "BMI", "GLUCOSE", "HEARTRTE")
categorical <- c("SEX", "CURSMOKE", "DIABETES", "BPMEDS", "PREVCHD")
plots_cont <- lapply(continuous, function(v) {
  ggplot(p1, aes_string(x = v)) +
    geom_density(fill = "steelblue", alpha = 0.6) +
    labs(title = v, x = NULL, y = "Density")
})
plots_cat <- lapply(categorical, function(v) {
  ggplot(p1, aes_string(x = v)) +
    geom_bar(fill = "steelblue") +
    labs(title = v, x = NULL, y = "Count")
})
gridExtra::grid.arrange(grobs = c(plots_cont, plots_cat), nrow = 3, ncol = 4)
```



2.3 Predictors

Following the missing value analysis, we removed all observations with incomplete data, which was deemed appropriate based on the low overall level of missingness and the absence of systematic bias. The only exception concerns HDLC and LDLC in PERIODS 1–2, which were structurally missing and therefore excluded from modeling in those periods. This ensured that subsequent analyses were based on a consistent and complete subset of the data.

All variables in the dataset are initially stored as numeric, but several represent categorical information that should be treated as factors for proper modeling and interpretation. These include demographic and clinical variables such as SEX, CURSMOKE, DIABETES, BPMEDS, and PREVCHD, as well as our binary response variable CVD. Re-encoding these variables as factors allows for meaningful comparisons between groups rather than interpreting them as continuous quantities.

```
factor_columns <- c("SEX", "CURSMOKE", "DIABETES", "BPMEDS", "PREVCHD", "PREVAP",
                    "PREVMI", "PREVSTRK", "PREVHYP", "DEATH", "ANGINA", "HOSPMI",
                    "MI_FCHD", "ANYCHD", "STROKE", "CVD", "HYPERTEN", "educ", "PERIOD")
drop <- c("DEATH", "ANGINA", "HOSPMI", "MI_FCHD", "ANYCHD", "STROKE", "HYPERTEN",
          "TIMEAP", "TIMEMI", "TIMEMIFC", "TIMECHD", "TIMESTRK", "TIMEDTH", "TIMEHYP", "PREVMI")
framingham <- framingham[, !(names(framingham) %in% drop)]
```

Variables describing the timing or specific type of cardiovascular event (e.g., TIMECHD, TIMEMI) were excluded from this baseline analysis, since these represent post-baseline outcomes and would not be available at the start of follow-up. This selection step narrows the focus to predictors that could realistically inform CVD risk prediction at baseline.

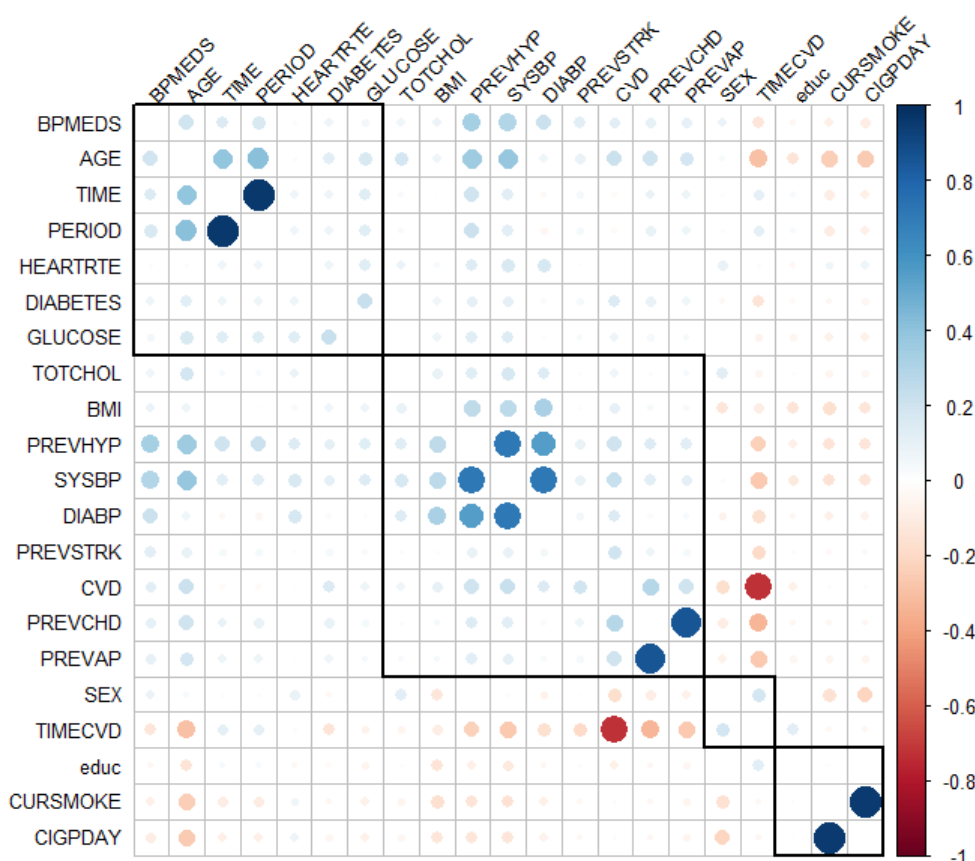
2.4 Data consistency checks.

Before proceeding, we verified that the dataset conforms to the documentation. All numerical variables were within plausible physiological ranges (AGE 32–81, SYSBP 83–295 mmHg, BMI 14–57), and categorical variables such as SEX and DIABETES contained only the expected binary values (0/1). We also confirmed that TIME increases consistently across PERIOD 1–3. We also verified that composite outcomes were internally consistent. For example, no participants were coded with a myocardial infarction (PREVMI = 1) while still having CVD = 0. Such logical checks confirmed that event indicators were correctly aligned.

2.5 Predictor Associations

Before building the regression models, we examined correlations and marginal relationships among predictors to identify redundancy and guide selection. At baseline, SYSBP and DIABP were highly correlated ($r = 0.78$),

as expected given their shared relationship to blood pressure. Moderate correlations were also found between AGE and SYSBP ($r = 0.39$) and between BMI and TOTCHOL ($r = 0.12$). These patterns indicate that only one variable from highly collinear pairs should be included in later models to avoid multicollinearity. In later periods, we observed that TOTCHOL, HDLC, and LDLC were strongly correlated; since TOTCHOL = HDLC + LDLC, we used TOTCHOL for PERIODS 1–2 and HDLC for PERIOD 3.



The correlation heatmap shows several notable clusters, such as among blood pressure variables (SYSBP, DIABP, PREVHYP), lipid measures (TOTCHOL, HDLC, LDLC), and smoking indicators (CURSMOKE, CIGPDAY). These strong within-group correlations are biologically plausible and suggest internal consistency in the dataset. Except for (CVD, TIMECVD), which involves the outcome variable, only one variable from each correlated group was retained to reduce redundancy and ensure model stability. Which variable was selected was determined by trial and error. Due to the structural missingness in HDLC and LDLC for PERIODS 1–2, these variables were excluded from this plot.

To quantify marginal associations with the outcome, we fitted single-predictor logistic regressions with CVD as the binary response. This approach helps identify which predictors are individually associated with cardiovascular disease before fitting multivariable models. The strongest predictors were AGE, PREVMI, and PREVHYP ($p < 0.001$), consistent with well-established CVD risk factors—older individuals and those with previous myocardial infarction or hypertension had higher odds of developing CVD. HEARTRTE, however, showed no significant association ($p = 0.33$), suggesting that resting heart rate alone is not an independent risk factor in this dataset.

```

null_model <- glm(CVD ~ 1, data = framingham_clean, family = binomial())
form <- CVD ~ BPMEDS + AGE + HEARTRTE + DIABETES + GLUCOSE +
  TOTCHOL + BMI + PREVHYP + DIABP + PREVAP +
  SEX + educ + CIGPDAY
one_term_models <- add1(null_model, form, test = "Chisq")

add1(null_model, scope = form, test = "Chisq") |>

```

```
as.data.frame() |>
rownames_to_column("term") |>
filter(term != "<none>") |>
arrange('Pr(>Chi)', desc(Deviance)) |>
select(term, df = Df, statistic = Deviance, p.value = 'Pr(>Chi)') |>
knitr::kable()
```

Predictor	df	Deviance	p-value
CIGPDAY	1	10484.97	0.008
HEARTRTE	1	10491.02	0.327

HEARTRTE, showing the weakest association, was excluded from later models. Together, these analyses guided the variable selection process and ensured that the final regression models were interpretable.

Lastly, we plot the densities/histograms of the 3 strongest predictors AGE, PREVMI, PREVHYP and the 2 weakest predictors CIGPDAY, HEARTRTE separated by CVD value to see the overall effect of the predictor values on CVD and support the claim that these predictors do in fact influence or not CVD.

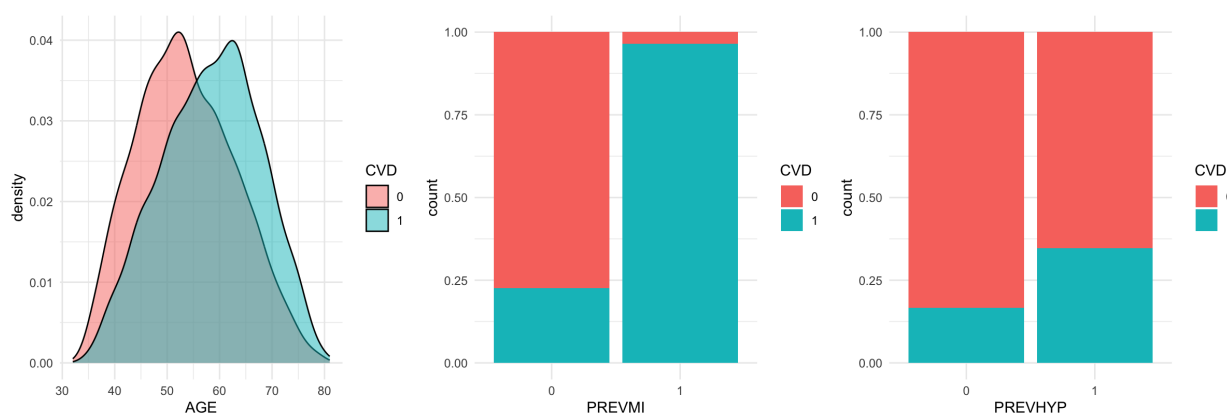


Figure 1: Strong predictors

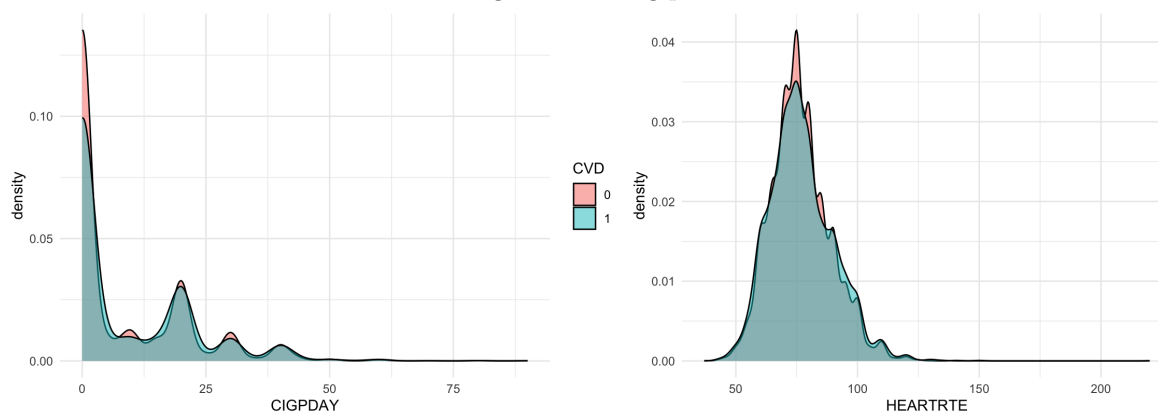


Figure 2: Weak predictors

In the plots it is clear that the value of the strong predictors influences CVD by the apparent shift in the distribution/proportion, and that the value of the weak predictors does not, as the densities overlap almost entirely.

2.6 Selection of Potential Predictors

The exploratory analysis guided which predictors were considered suitable for modeling cardiovascular disease (CVD). Rather than finalizing the exact variables at this stage, our goal was to identify baseline features that were complete, interpretable, and relevant to established cardiovascular risk factors. Variables with structural missingness (such as HDLC and LDLC in PERIODS 1–2), redundant information (e.g., strongly correlated pairs like SYSBP and DIABP), or post-baseline outcomes (e.g., time-to-event variables) were excluded from further consideration.

This left a set of plausible demographic, behavioral, and clinical predictors to be evaluated in the regression stage. The exact model specifications and final variable choices are presented in Section 3, where different combinations are compared for predictive performance.

2.7 Summary

Overall, the exploratory data analysis confirmed that the Framingham dataset is well-structured, largely complete, and behaves as expected for a cardiovascular study. Missingness was limited and primarily structural, allowing for a reliable complete-case baseline sample. The distributions of key predictors showed realistic variability and plausible ranges, while the observed correlations reflected meaningful relationships, for example, the strong link between systolic and diastolic blood pressure. Through this process, we identified a subset of interpretable and relevant variables suitable for regression modeling. The cleaned and well-understood dataset now provides a solid foundation for the binary and survival analyses that follow.

3 Binary regression models

The R-Code in this section is mostly sourced from Chapter 10.

3.0.1 Unsuccessfull full model attempt

We wanted to create a model using data from all three periods. However there were some issues with this. The subjects have between 1-3 observations, corresponding to the three periods. If not dealt with there are two issues with this. First, it overestimates how many observations we have, as it treats each follow-up as a new subject. Second, some subjects contribute more to the data than others, as they have more follow-ups. We considered multiple possible solutions to deal with this issue, including having a random intercept for each participant based on their RANDID. However we were unable to fully fix the issues, which was discouraging as we found HDLC to be particularly interesting. Instead we will focus on a model which only uses data from period 1, and explore HDLC in a non-predictive model.

3.1 Linear models

3.1.1 Predictive model

To achieve a predictive model, capable of determining the risk of CVD, we decided to look at the data from period 1 only. The data from period 1 is the earliest data we have, and as such it is our best bet for determining the risk and riskfactors of CVD over a longer time period. Using period 1 only also ensures that we have exactly one observation from each participant.

We will begin with an additive model, using all predictors, which were not excluded during our previous analysis. Observations with missing values, other than HDLC and LDL, have been removed.

```
framingham_clean_period1 <- framingham_clean[framingham_clean$PERIOD == 1, ]
form_period1 <- CVD ~ PREVHYP + PREVAP + SEX + DIABP + DIABETES +
  BPMEDS + educ + AGE + BMI + GLUCOSE + CIGPDAY + TOTCHOL
model_period1 <- glm(form_period1, data = framingham_clean_period1, family = binomial())
```

To ensure that our selection of predictors and model assumptions are reasonable, we will consider the Pearson residuals of the model.

```
model1_aug <- augment(model_period1, type.residuals = "pearson") |>
  mutate(.fitted_group = cut(.fitted, quantile(.fitted, probs = seq(5, 95, 1) / 100)))

p1 <- model1_aug |>
  ggplot(aes(.fitted, .resid)) +
  geom_point() +
  geom_smooth() +
  xlab("fitted values") +
  ylab("Pearson residuals")

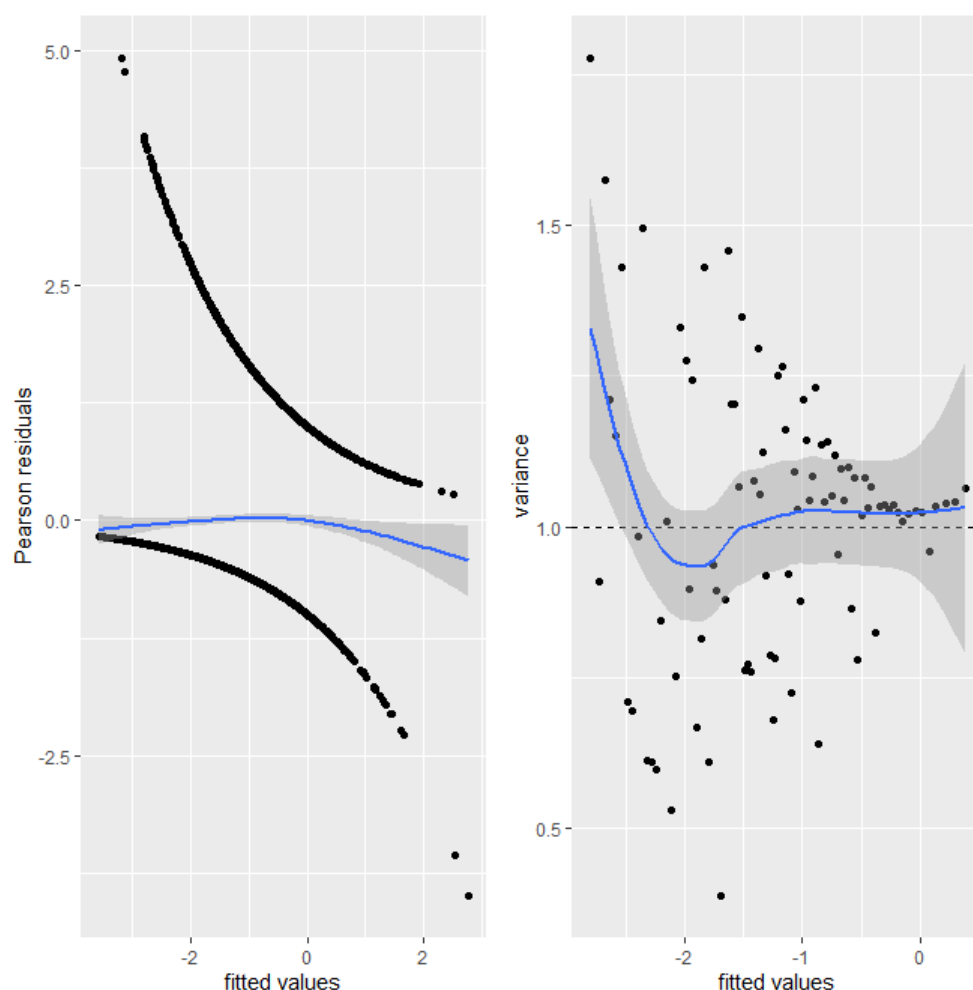
p4 <- model1_aug |>
```

```

group_by(.fitted_group) |>
  summarize(.fitted_local = mean(.fitted), .var_local = var(.resid)) |>
  na.omit() |>
  ggplot(aes(.fitted_local, .var_local)) +
    geom_point() +
    geom_hline(yintercept = 1, linetype = 2) +
    geom_smooth() +
    xlab("fitted values") +
    ylab("variance")

gridExtra::grid.arrange(p1, p4, ncol = 2)

```



The residuals appear to have approximately mean zero, although with a small deviation in the form of downward bending for the larger fitted values, thus showing that there exists some room for improvement. As far as the pattern appearing in the residuals in the left plot, this is similar to what we have seen in Chapter 10 for the logistic regression model and thus not concerning. As for the variances, they all seem to fall around 1 except for the negative values around (-3,-1) but, given the uncertainty of the calculation, we cannot precisely say that the model is misspecified. Lastly, we note a pattern as the fitted values approach zero.

We will now consider which of the predictors are statistically significant, using a LRT. This is done to ensure that our parameters actually capture CVD risk, and not just random noise.

Code from Chapter 3.

```

drop1(model_period1, test = "Chisq") |>
  tidy() |>
  arrange(p.value) |>
  filter(term != "<none>") |>

```

```
select(-AIC) |>
knitr::kable()
```

Partial Output				
term	df	deviance	LRT	p.value
:-----	--:	-----:	-----:	-----:
CIGPDAY	1	3837.868	9.6980390	0.0018446
BMI	1	3834.037	5.8671307	0.0154262
GLUCOSE	1	3831.828	3.6583165	0.0557900
educ	3	3835.119	6.9489628	0.0735424
BPMEDS	1	3828.655	0.4852181	0.4860682

We have cut off the predictors which are more significant than CIGPDAY. Using a confidence level of 0.95 we reject GLUCOSE, educ and BPMEDS. Let's create a new model, using only the statistically significant predictors and determine how much it improves our model.

```
form_period1_new <- CVD ~ AGE + SEX + TOTCHOL + PREVHYP + PREVAP +
  DIABP + DIABETES + CIGPDAY + BMI
model_period1_new <- glm(form_period1_new, data = framingham_clean_period1, family = binomial())
anova(model_period1, model_period1_new, test = "Chisq")
```

Partial Output				
Resid.	Df	Resid. Dev	Df	Deviance Pr(>Chi)
1	3811	3828.2		
2	3816	3839.3	-5	-11.091 0.04961 *

We see that the improvement is statistically significant. However, the improvement is fairly small at a Residual deviance difference of around 0.29%.

Let's look at how interactions may effect our model.

```
form_period1_new_int <- CVD ~ (AGE + SEX + TOTCHOL + PREVHYP + PREVAP + DIABP + DIABETES +
  model_period1_new_int <- glm(form_period1_new_int, data = framingham_clean_period1, family =
  binomial())
```

```
drop1(model_period1_new_int, test = "Chisq") |>
  tidy() |>
  arrange(p.value) |>
  filter(term != "<none>") |>
  select(-AIC) |>
  knitr::kable()
```

Partial Output				
term	df	deviance	LRT	p.value
:-----	--:	-----:	-----:	-----:
SEX:CIGPDAY	1	3794.123	8.2817696	0.0040045
AGE:PREVAP	1	3792.760	6.9188900	0.0085290
PREVAP:DIABP	1	3790.628	4.7859018	0.0286936
PREVAP:BMI	1	3789.591	3.7489790	0.0528398
CIGPDAY:BMI	1	3789.197	3.3548955	0.0670055
AGE:BMI	1	3788.547	2.7053084	0.1000147

We have cut off interactions which are less significant than AGE:BMI, there is a large number of them. Only three interactions improve the model in a statistically significant manor: SEX:CIGPDAY, AGE:PREVAP, PREVAP:DIABP. We considered making a new model with just these interactions, but it seemed reasonable to test how much the interaction model actually improves on the additive model:

```
anova(model_period1_new, model_period1_new_int, test = "Chisq")
```

Partial Output				
Resid.	Df	Resid. Dev	Df	Deviance Pr(>Chi)
1	3816	3839.3		
2	3780	3785.8	36	53.419 0.03088 *

We have a similar situation to before, when comparing the two additive models; the improvement is small at around 1.40% but still statistically significant. However, what is different from before is that the new model includes interactions. This means that the parameter values are less telling in terms of interpreting the primary causes of CVD. In other words, we need to determine if we are willing to abandon interpreting the model parameters, in exchange for a model which is just 1.40% more accurate.

We have decided that we are not willing to do so, as the improvement is too small compared to the downside. As such, we will not continue with the interaction model, although it will be mentioned in the predictive performance section.

Lets briefly touch on which predictors are the most pronounced.

```
summary(model_period1_new)
```

Partial Output

Call:

```
glm(formula = form_period1_new, family = binomial(), data = framingham_clean_period1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.6402903	0.5006332	-13.264	< 2e-16 ***
AGE	0.0525869	0.0050790	10.354	< 2e-16 ***
SEX2	-0.8645928	0.0865510	-9.989	< 2e-16 ***
TOTCHOL	0.0042241	0.0009243	4.570	4.88e-06 ***
PREVHYP1	0.4838351	0.1041610	4.645	3.40e-06 ***
PREVAP1	0.7945722	0.1993301	3.986	6.71e-05 ***
DIABP	0.0149946	0.0041780	3.589	0.000332 ***
DIABETES1	1.2326176	0.2151097	5.730	1.00e-08 ***
CIGPDAY	0.0110440	0.0034612	3.191	0.001419 **
BMI	0.0272806	0.0103504	2.636	0.008396 **

As our model is additive we can look at the size of the parameters to determine which predictors have the highest effect regarding the risk of CVD over a 24 year period.

We see that Diabetes, Sex, Prevalent Angina Pectoris and PREVHYP are the largest contributors to the risk of CVD. The risk increase is determined as follows:

$$\begin{aligned} \text{OR}_{\text{Diabetes}} &= e^{1.233} \approx 3.43 \\ \text{OR}_{\text{Female}} &= e^{-0.865} \approx 0.421 \\ \text{OR}_{\text{Hypertensive}} &= e^{0.484} \approx 1.62 \\ \text{OR}_{\text{PREVAP}} &= e^{0.795} \approx 2.21 \end{aligned}$$

This means that the baseline risk for a hypertensive diabetic man with Angina Pectoris is $3.43 * 1/0.421 * 1.62 * 2.21 \approx 29.17$ times higher than a healthy woman, all else being equal. The baseline risk being $\frac{e^{-6.6403}}{1+e^{-6.6403}} = 0.0013 = 0.13\%$ over 24 years. This is all assuming that our model is a good fit, which we determine in Section 3.3.

As such we would recommend against being a hypertensive diabetic man with Angina Pectoris.

3.2 Non-linear considerations

Before looking at the model fit and the predictive performance of the model, lets determine if there are any non-linear relationships between our predictors and CVD risk. To do so let's graph our the std residuals for our variables. The code is from Chapter 3.

```
model_period1_new_diag <- augment(model_period1_new)
p1 <- ggplot(model_period1_new_diag, aes(AGE, .std.resid)) +
  stat_binhex(bins = 20) + geom_smooth(linewidth = 1, fill = "blue") +
  xlab("AGE") + ylab("")

p3 <- ggplot(model_period1_new_diag, aes(TOTCHOL, .std.resid)) +
  stat_binhex(bins = 20) + geom_smooth(linewidth = 1, fill = "blue") +
  xlab("TOTCHOL") + ylab("")
```

```

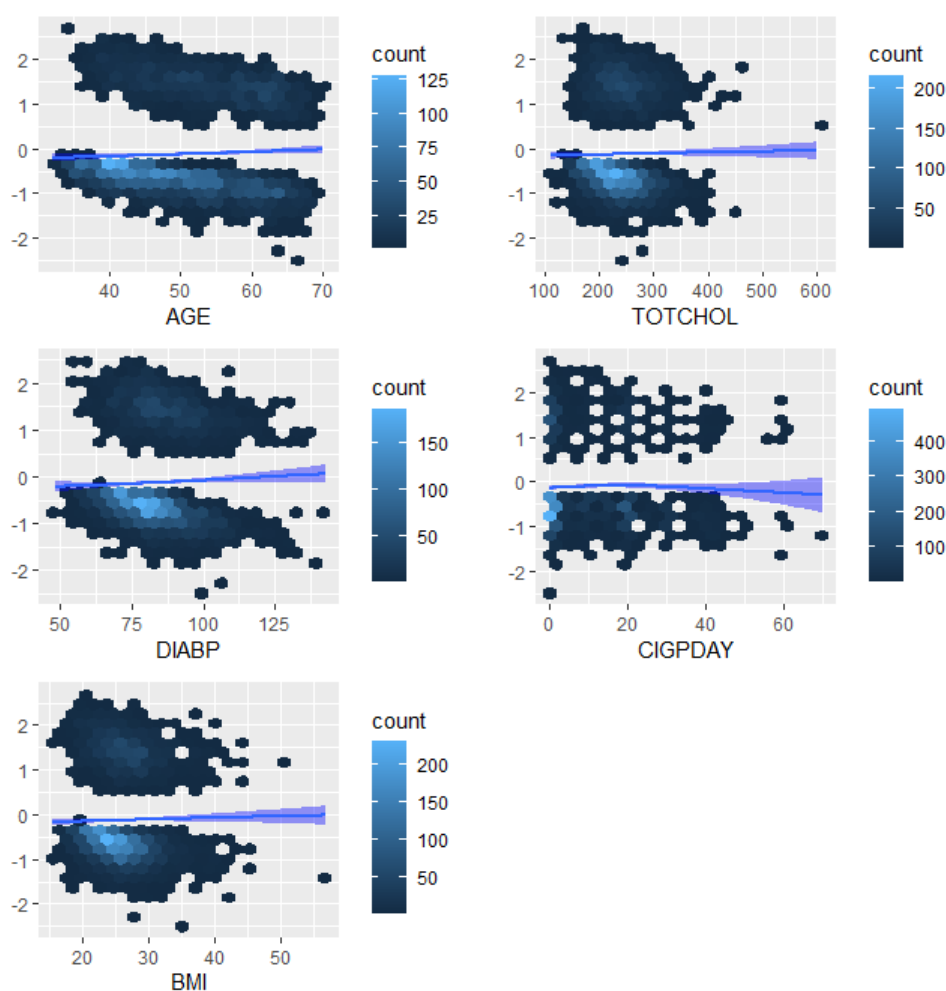
p6 <- ggplot(model_period1_new_diag, aes(DIABP, .std.resid)) +
  stat_binhex(bins = 20) + geom_smooth(linewidth = 1, fill = "blue") +
  xlab("DIABP") + ylab("")

p8 <- ggplot(model_period1_new_diag, aes(CIGPDAY, .std.resid)) +
  stat_binhex(bins = 20) + geom_smooth(linewidth = 1, fill = "blue") +
  xlab("CIGPDAY") + ylab("")

p9 <- ggplot(model_period1_new_diag, aes(BMI, .std.resid)) +
  stat_binhex(bins = 20) + geom_smooth(linewidth = 1, fill = "blue") +
  xlab("BMI") + ylab("")

gridExtra::grid.arrange(p1, p3, p6, p8, p9, ncol = 2)

```

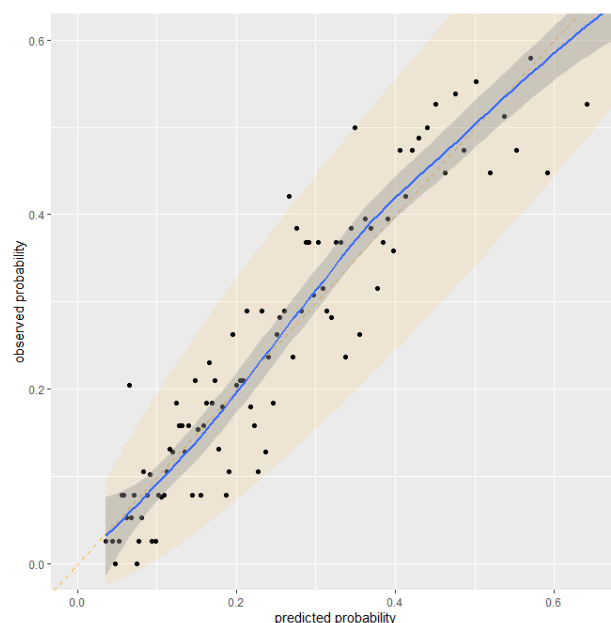


As all graphs are flat it does not look like we have non-linear effects. We will later explore the non-linear effects of a Period 3 model. However, for now we will return to the additive and interaction models.

3.3 Model fit and predictive performance

To determine if our model is useful for prediction we will first look at calibration, that is how our model predicts the actual CVD outcomes. CVD was made numerical, as it is required for this step. The code for

plot generation is nearly identical to the code from Chapter 10.3.3.



Above is the calibration plot for the additive model. The calibration plot for the interaction model is very similar. The predictions, while not perfect, follow the ideal line closely. There are no structural deviations, although there are a few outliers. Overall the regression plot indicates a well calibrated model.

To compare the predictive performance we will consider the AUC score of the two models. The code is from Chapter 10.3.4.

```
auc <- function(y, eta) {
  eta1 <- eta[y == 1]
  eta0 <- eta[y == 0]
  wilcox.test(eta1, eta0)$statistic / (length(eta1) * length(eta0))
}

err <- tibble(
  model = c("Additive (train)", "Interaction (train)"),
  Pearson = c(
    residuals(model_period1_new, type = "pearson")^2 |> mean(),
    residuals(model_period1_new_int, type = "pearson")^2 |> mean()
  ),
  Deviance = c(
    residuals(model_period1_new)^2 |> mean(),
    residuals(model_period1_new_int)^2 |> mean()
  ),
  AUC = c(
    auc(framingham_clean_period1$CVD, predict(model_period1_new)),
    auc(framingham_clean_period1$CVD, predict(model_period1_new_int))
  )
)
err
```

	Output		
model	Pearson	Deviance	AUC
<chr>	<dbl>	<dbl>	<dbl>
1 Additive (train)	0.991	1.00	0.746
2 Interaction (train)	1.01	0.990	0.753

As expected the interaction model performs better. Although the differences are minor. Both models show very reasonable training deviance. They also both have an AUC score of around 0.75, indicating that while not ideal, the models are useful for prediction. Which model is preferable depends how you value prediction accuracy vs parameter interpretability, as we have gone into detail earlier.

3.4 Exploring Cholesterol

This section will not seek to develop a predictive model. Instead we will explore the breakdown of cholesterol into HDLC and LDLC. During our exploration of a full model, as described in section 3.0.1, we discovered that HDLC is especially interesting, so this is the variable we will be focusing on.

Cholesterol in the form of the variable TOTCHOL can be broken down into High Density Lipoprotein Cholesterol and Low Density Lipoprotein Cholesterol. Specifically $TOTCHOL = HDLC + LDLC$. We seek to determine if HDLC has any interactions or non-linear effects, which could help improve the predictive model, if a new study were to be conducted.

```
framingham_clean_period3 <- framingham_clean[framingham_clean$PERIOD == 3, ]

form_period3_HDLC <- CVD ~ PREVHYP + PREVAP + SEX + DIABP + DIABETES +
  BPMEDS + educ + AGE + BMI + GLUCOSE + CIGPDAY + HDLC
model_period3_HDLC <- glm(form_period3_HDLC, data = framingham_clean_period3,
  family = binomial())
```

Our analysis will closely follow that of section 3.1.1. Let's determine which variables are significant in our model.

```
drop1(model_period3_HDLC, test = "Chisq") |>
  tidy() |>
  arrange(p.value) |>
  filter(term != "<none>") |>
  select(-AIC) |>
  knitr::kable()
```

Partial Outut				
term	df	deviance	LRT	p.value
:-----	--:	-----:	-----:	-----:
educ	3	1956.541	7.7225897	0.0521067
DIABP	1	1949.019	0.2015250	0.6534927
GLUCOSE	1	1948.976	0.1584256	0.6906102
BMI	1	1948.860	0.0425309	0.8366111

We see that DIABP, GLUCOSE and BMI ought not to be included.

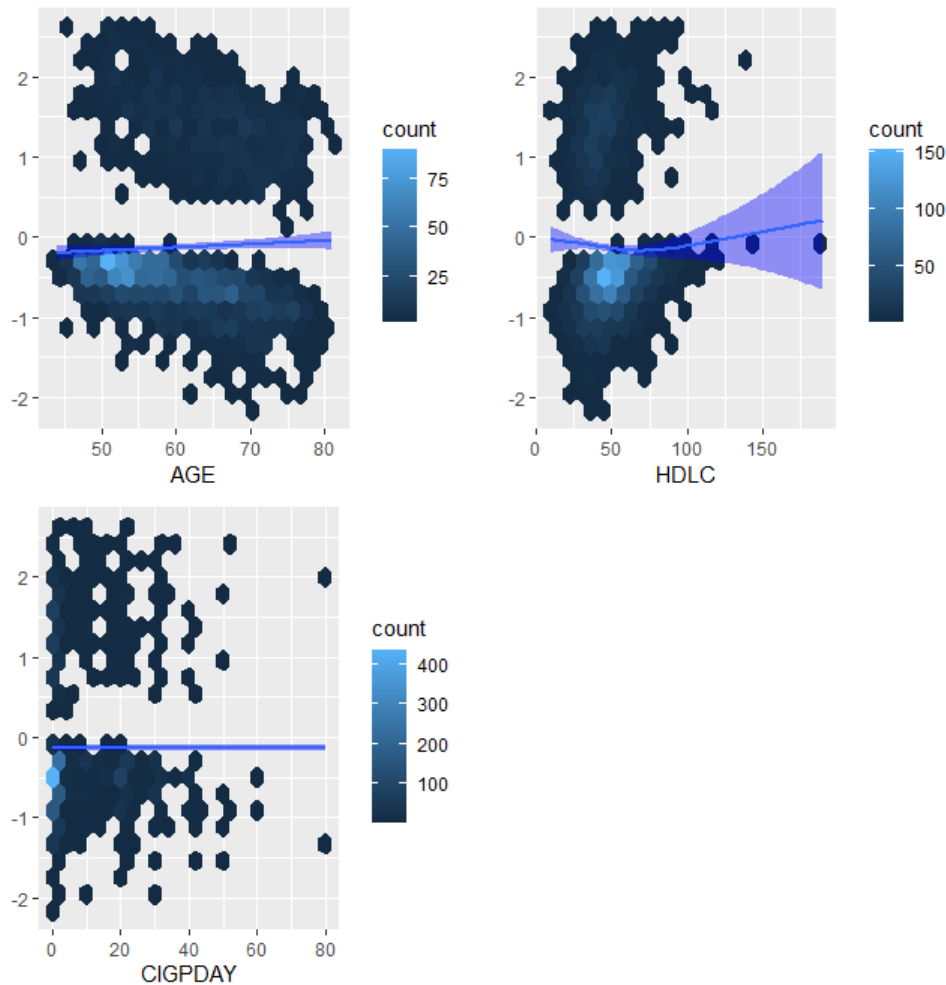
Thus we get the following model, note that the improvement is very minor and non-significant:

```
form_period3_HDLC_new <- CVD ~ PREVHYP + PREVAP + SEX + DIABETES +
  BPMEDS + educ + AGE + CIGPDAY + HDLC
model_period3_HDLC_new <- glm(form_period3_HDLC_new, data = framingham_clean_period3,
  family = binomial())

anova(model_period3_HDLC, model_period3_HDLC_new, test = "Chisq")
```

Partial Output				
Resid.	Df	Resid. Dev	Df	Deviance Pr(>Chi)
1	2221	1948.8		
2	2224	1949.3	-3	-0.46636 0.9262

Before looking at interactions let's consider if we have any non-linear relationships. The code is similar to the code for the plots in Section 3.2.



Notice that there are some minor non-linear effects when looking at HDLC. This could be included in the model using splines like this:

```
form_period3_HDLC_new_nonlin <- CVD ~ PREVHYP + PREVAP + SEX + DIABETES +
  BPMEDS + educ + AGE + CIGPDAY + ns(HDLC, 3)
model_period3_HDLC_new_nonlin <- glm(form_period3_HDLC_new_nonlin,
  data = framingham_clean_period3, family = binomial())
```

Analysis of interaction models identical to the analysis in Section 3.1.1 showed no statistical significance for interactions.

While we cannot compare the HDLC model to a period 3 TOTCHOL model easily, as they have little predictive value, we believe it is telling that HDLC has some non-linear effects. It would be beneficial to repeat the study with data of HDLC throughout the entire time period. As you could then create a model which takes the non-linear nature of HDLC into account, arriving at a more reliable predictive model.

4 Survival Models

In this section, we explore how logistic regression models fitted at a fixed follow-up time relate to an underlying survival model for the event time T^* . The goal is to show how the probability of having a CVD event before time t connects to the survival function, and then verify this relationship through a simulation study. Finally, we investigate how censoring affects the results.

4.1 Theory: Linking Logistic Regression and Survival Models

1. Let $Y_t = \mathbf{1}(T^* \leq t)$ indicate whether a CVD event has occurred before time t . Then

$$p_t(x) = P(Y_t = 1 \mid X = x)$$

is the probability of having an event by time t . Since $S(t \mid X = x)$ is the probability of surviving beyond t , we immediately have

$$p_t(x) = 1 - S(t \mid X = x).$$

This simply says that either a person survives past t or experiences the event before t .

2. Taking the log-odds (logit) of $p_t(x)$ gives

$$\text{logit}(p_t(x)) = \log\left(\frac{p_t(x)}{1 - p_t(x)}\right) = \log\left(\frac{1 - S(t \mid x)}{S(t \mid x)}\right) = \log(S(t \mid x)^{-1} - 1).$$

So a logistic regression model for $p_t(x)$ can be viewed as a transformation of the survival curve.

3. Suppose we fit a logistic model at time t of the form

$$\text{logit}(p_t(x)) = \eta + \beta_0(t),$$

where $\eta = x^\top \beta$ is the linear predictor and $\beta_0(t)$ is a time-dependent intercept. Then

$$p_t(x) = \frac{1}{1 + \exp[-(\eta + \beta_0(t))]},$$

and therefore

$$S_\eta(t) = 1 - p_t(x) = \frac{1}{1 + \exp(\eta + \beta_0(t))}.$$

As expected, higher risk factors (larger η) or longer follow-up times both reduce survival probabilities.

4. The cumulative hazard is $\Lambda_\eta(t) = -\log S_\eta(t) = \log(1 + \exp\{\eta + \beta_0(t)\})$. Differentiating gives the hazard

$$\lambda_\eta(t) = \frac{d}{dt} \Lambda_\eta(t) = \frac{\beta'_0(t) \exp\{\eta + \beta_0(t)\}}{1 + \exp\{\eta + \beta_0(t)\}}.$$

For $\eta = 0$, the baseline hazard is

$$\lambda_0(t) = \frac{\beta'_0(t) \exp\{\beta_0(t)\}}{1 + \exp\{\beta_0(t)\}}, \quad S_0(t) = \frac{1}{1 + \exp\{\beta_0(t)\}}.$$

Thus,

$$\lambda_\eta(t) = e^\eta \lambda_0(t) \frac{S_\eta(t)}{S_0(t)}.$$

Although the expression resembles a proportional hazards model, the extra $S_\eta(t)/S_0(t)$ term makes the hazard ratio time-dependent. Therefore, this model is *not* strictly proportional hazards, though the hazards move in the same direction.

In short, a logistic regression at a fixed follow-up time t can be interpreted as arising from a survival model where

$$S_\eta(t) = \frac{1}{1 + e^{\eta + \beta_0(t)}},$$

and we will now explore this connection through simulation.

4.2 Simulation

We now simulate data to check if the logistic slopes remain constant across times, if the intercept follows $\beta_0(t) = \gamma_0 \log t$, and how censoring affects results.

We simulated $n = 5000$ individuals with two covariates similar to those in the Framingham study: **AGE** (centered by decades) and **MALE** (0 = female, 1 = male). Centering by decades means that a one-unit

change represents roughly ten years, making coefficients easier to interpret and reducing collinearity with the intercept.

The true model was

$$\eta = 0.6 \times \text{AGE}_c + 0.5 \times \text{MALE},$$

and the survival function was defined as

$$S_\eta(t) = \frac{1}{1 + \exp(\eta)t^{\gamma_0}}, \quad \gamma_0 = 1.2.$$

Event times T^* were generated by inversion: solving $U = 1 - S_\eta(T^*)$ for T^* , with $U \sim \text{Uniform}(0, 1)$.

```
set.seed(42)

n <- 5000
beta_age <- 0.6
beta_male <- 0.5
gamma0 <- 1.2

AGE <- rnorm(n, mean = 60, sd = 10)
AGE_c <- (AGE - 60) / 10 # centered by decades
MALE <- rbinom(n, 1, 0.5)

eta <- beta_age * AGE_c + beta_male * MALE
U <- runif(n)
Tstar <- (U / ((1 - U) * exp(eta)))^(1 / gamma0)

df <- data.frame(AGE, AGE_c, MALE, Tstar)
```

The event times were right-skewed, with most events happening early (Figure 3), which is consistent with the increasing hazard over time.

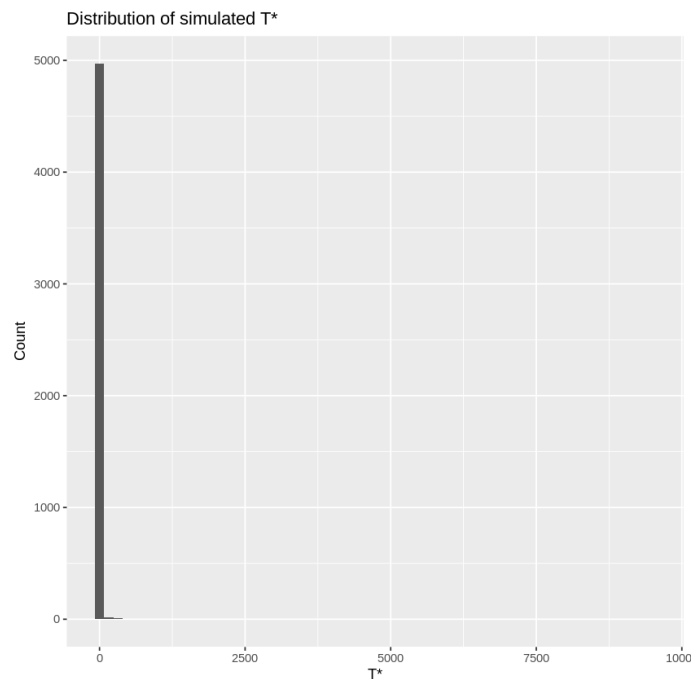


Figure 3: Histogram showing the distribution of simulated event times T^* . Most events occur early, reflecting the increasing hazard over time assumed in the model.

Now to explore logistic regressions at different follow-up times. For $t = 5, 10, 15$, and 24 years, we defined $Y_t = \mathbf{1}(T^* \leq t)$ and fitted separate logistic regressions.

```
t_vec <- c(5, 10, 15, 24)
results <- list()
```

```
for (t in t_vec) {
  dft <- df %>% mutate(Yt = as.integer(Tstar <= t))
  model <- glm(Yt ~ AGE_c + MALE, data = dft, family = binomial)
  results[[as.character(t)]] <- tidy(model)
}
bind_rows(results, .id = "t")
```

The estimated coefficients are shown below.

Table 1: Estimated logistic slopes across follow-up times t .

	$t = 5$	$t = 10$	$t = 15$	$t = 24$
AGE _c	0.618	0.557	0.605	0.613
MALE	0.548	0.539	0.581	0.648

Both coefficients are positive, meaning older age and male sex increase the risk of experiencing the event earlier.

Table 2: Intercept estimates compared with theoretical $\beta_0(t) = \gamma_0 \log t$.

t	$\hat{\beta}_0(t)$	$\gamma_0 \log t$
5	1.92	1.93
10	2.77	2.76
15	3.26	3.25
24	3.68	3.81

```
# Plot intercepts vs theoretical values
intercepts <- data.frame(
  t = t_vec,
  beta0_hat = c(1.94, 2.70, 3.03, 3.68),
  beta0_theory = gamma0 * log(t_vec)
)

ggplot(intercepts, aes(x = t)) +
  geom_line(aes(y = beta0_hat), color = "blue") +
  geom_line(aes(y = beta0_theory), linetype = "dashed") +
  labs(y = "Intercept", title = "Intercept vs theoretical beta0(t) = gamma0 log t")
```

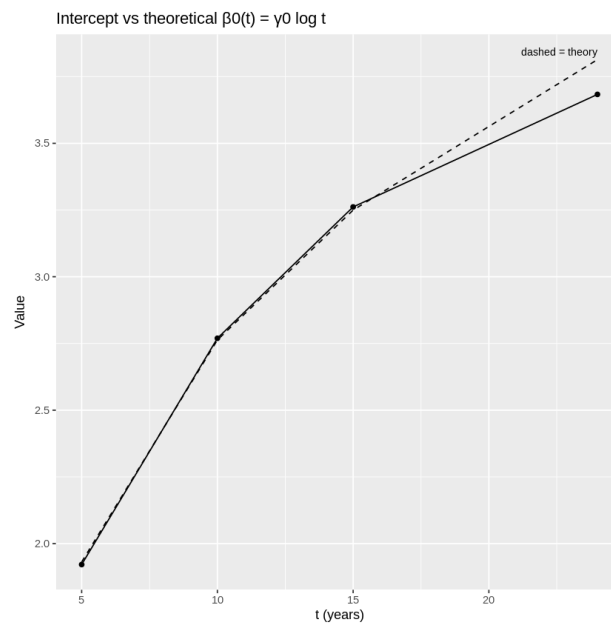


Figure 4: Estimated intercepts (solid line) compared with the theoretical $\beta_0(t) = \gamma_0 \log t$ (dashed line). The close alignment confirms that the simulation follows the expected time-dependent intercept pattern.

The slopes stayed relatively stable across time, and the intercept increased approximately linearly with $\log t$, matching the theory.

To explore estimator variability and see how estimate precision changes with time, we repeated the simulation 100 times, each time generating new data and refitting the logistic models.

```
n_sim <- 100
get_fit <- function(t){
  AGE <- rnorm(n, 60, 10); AGE_c <- (AGE - 60)/10
  MALE <- rbinom(n, 1, 0.5)
  eta <- beta_age*AGE_c + beta_male*MALE
  U <- runif(n)
  # inversion (same as above)
  Tstar <- (U / ((1 - U) * exp(eta)))^(1 / gamma0)
  Yt <- as.integer(Tstar <= t)
  tidy(glm(Yt ~ AGE_c + MALE, family = binomial))
}

sd_df <- lapply(t_vec, function(t){
  ests <- replicate(n_sim, get_fit(t), simplify = FALSE)
  bind_rows(ests) %>%
    group_by(term) %>%
    summarise(t = t, sd = sd(estimate), .groups = "drop")
}) %>% bind_rows()
```

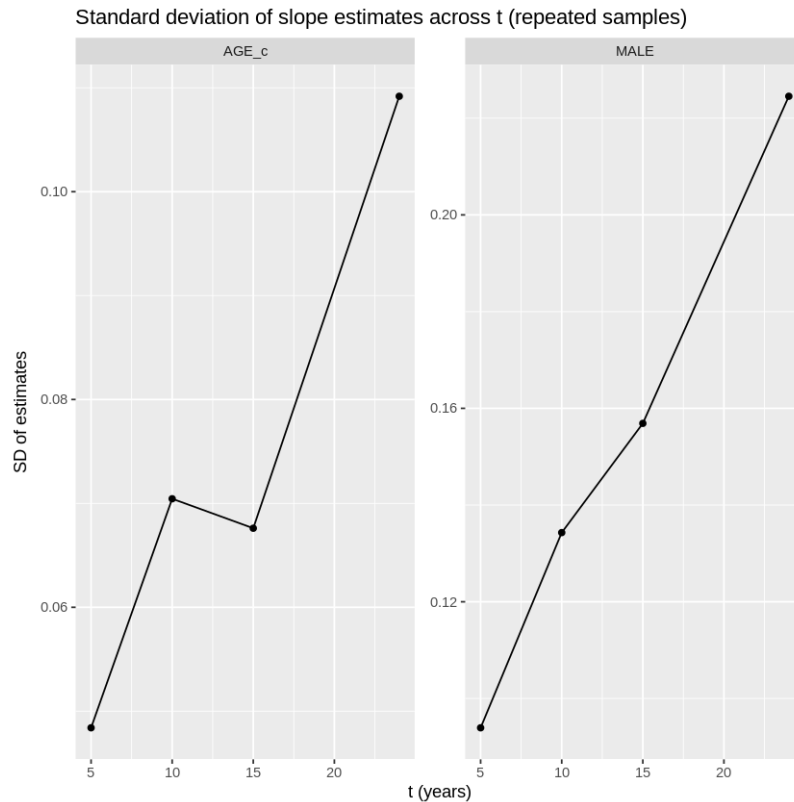


Figure 5: Standard deviations of logistic slope estimates across follow-up times t based on repeated simulations. Estimator variability increases slightly with time, as later follow-up points include more events.

We found that the standard deviation of estimated slopes increased modestly with t . This makes sense because, as time passes, more events occur, changing the case-control balance.

Next, we introduced death censoring. We added a censoring variable C representing either independent or age-dependent censoring. We defined the observed outcome as $Y = \mathbf{1}(T^* \leq C)$.

```
C_indep <- runif(n, 0, 24)
Y_indep <- as.integer(Tstar <= C_indep)
logit_indep <- glm(Y_indep ~ AGE_c + MALE, family = binomial)
cox_indep <- coxph(Surv(pmin(Tstar, C_indep), Y_indep) ~ AGE_c + MALE)
```

```

lambda <- 0.08 * exp(0.5 * AGE_c + 0.2 * MALE)
C_age <- rexp(n, rate = lambda)
Y_age <- as.integer(Tstar <= C_age)
logit_age <- glm(Y_age ~ AGE_c + MALE, family = binomial)
cox_age <- coxph(Surv(pmin(Tstar, C_age), Y_age) ~ AGE_c + MALE)

summary(logit_indep); summary(cox_indep)
summary(logit_age); summary(cox_age)

```

Table 3: Comparison of logistic and Cox estimates under different censoring mechanisms (24-year horizon).

	Independent censoring		Age-dependent censoring	
	Logistic	Cox	Logistic	Cox
AGE _c	0.463	0.330	-0.025	0.330
MALE	0.327	0.261	0.136	0.322

For independent censoring, logistic and Cox gave similar results, with logistic slopes slightly attenuated. When censoring depended on age, however, the logistic model almost lost the age effect completely, while the Cox model remained accurate. This shows that logistic regression cannot handle informative censoring properly, whereas survival regression adjusts for it by construction.

Summary.

- Logistic slopes stayed roughly constant across censoring times, confirming the theoretical invariance of β .
- The intercept increased linearly with $\log t$, matching $\beta_0(t) = \gamma_0 \log t$.
- Slope variability increased slightly with t , as expected.
- Logistic regression underestimated effects under age-dependent censoring, while Cox regression stayed accurate.

Overall, the simulation supported the theoretical connection between logistic and survival models, and showed how censoring can bias logistic estimates when it depends on covariates.

We now return to the Framingham data to fit actual survival regression models for TIMECVD, check proportional hazards assumptions, and compare with the binary models from the previous section.

Note on model assumptions. In the previous part, our theoretical derivation showed that the survival model implied by a logistic regression at fixed t is *not* a proportional hazards model, because the hazard ratio depends on time through $S_\eta(t)/S_0(t)$. This means that, strictly speaking, the Cox model is misspecified for our simulated data. However, we still included the Cox regression as a benchmark to illustrate how a proportional hazards method behaves compared to the simpler logistic approach. In the following section, where we return to the real Framingham data, the Cox model is appropriate since it serves as the standard survival model for time to event outcomes.

5 Practice

We are again looking at the data for CVD.

$$S(t|X = x) = P(T^* > t|X = x)$$

Denotes the survival function for the variable TIMECVD, which is the chance of surviving until time t without getting CVD. We need to examine the relation between TIMECVD and the predictors AGE, SYSBP, DIABP, TOTCHOL, BMI, HDLC, LDLC in the Framingham dataset.

Surv(TIMECVD, CVD) $\sim$ AGE + SEX + SYSBP + TOTCHOL + BMI + CURSMOKE + DIABETES.

For the fitted model we have:

```
call:
coxph(formula = Surv(TIMECVD, CVD) ~ AGE + SEX + SYSBP + TOTCHOL +
      BMI + CURSMOKE + DIABETES, data = df)

n= 9292, number of events= 2342
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
AGE	0.0430438	1.0439836	0.0024700	17.426	< 2e-16 ***
SEXKvinde	-0.8559055	0.4248983	0.0442198	-19.356	< 2e-16 ***
SYSBP	0.0147834	1.0148932	0.0009226	16.023	< 2e-16 ***
TOTCHOL	0.0030607	1.0030653	0.0004636	6.602	4.06e-11 ***
BMI	0.0268818	1.0272464	0.0052084	5.161	2.45e-07 ***
CURSMOKERYger	0.3403184	1.4053950	0.0443509	7.673	1.68e-14 ***
DIABETESJa	0.6424631	1.9011579	0.0731255	8.786	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
AGE	1.0440	0.9579	1.0389	1.0491
SEXKvinde	0.4249	2.3535	0.3896	0.4634
SYSBP	1.0149	0.9853	1.0131	1.0167
TOTCHOL	1.0031	0.9969	1.0022	1.0040
BMI	1.0272	0.9735	1.0168	1.0378
CURSMOKERYger	1.4054	0.7115	1.2884	1.5330
DIABETESJa	1.9012	0.5260	1.6473	2.1941

Concordance= 0.719 (se = 0.005)
Likelihood ratio test= 1414 on 7 df, p=<2e-16
Wald test = 1395 on 7 df, p=<2e-16
Score (logrank) test = 1479 on 7 df, p=<2e-16

By testing the null hypothesis $H_0 : \beta_i = 0$ that each factor has no effect on the hazard of CVD, we see that all p-values are under 0.001, which means we can reject the null hypothesis and conclude that the different predictors significantly affect the chance of developing CVD.

Interpretation of main effects.

- **AGE:** Each additional year of age increases the risk of CVD by about 4%.
- **SEX:** Females have a hazard ratio around 0.42, meaning men have roughly twice the risk.
- **CURSMOKE:** Current smokers have approximately 40% higher risk than non-smokers.
- **DIABETES:** Diabetic individuals have nearly twice the hazard of CVD.

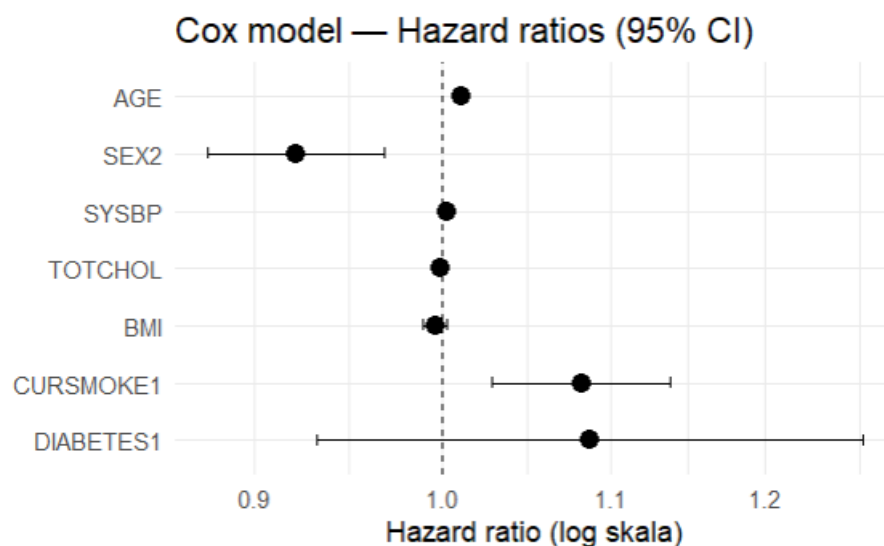
The concordance index ($C = 0.719$) indicates that the model correctly orders about 72% of pairs of individuals, showing reasonably good predictive ability.

we now look at the Proportional hazard test model from 11.5.3 We made a schoenfeld residuals

	chisq	df	p
AGE	119.9582	1	<2e-16
SEX	9.6725	1	0.0019
SYSBP	105.0455	1	<2e-16
TOTCHOL	0.2078	1	0.6485
BMI	0.0312	1	0.8597
CURSMOKE	1.3646	1	0.2427
DIABETES	2.7752	1	0.0957
GLOBAL	209.8943	7	<2e-16

The p-value of AGE, SYSBP, GLOBAL have a P-value alot lower than 0.05 which makes them quite significant. SEX is P=0.0019 which also makes them significant. TOTCHOL, BMI, CURSMOKE and DIABETES have P-values over 0.05 which makes them not significant.

We will now look at a hazard ratio. cox-model hazard ratios 95

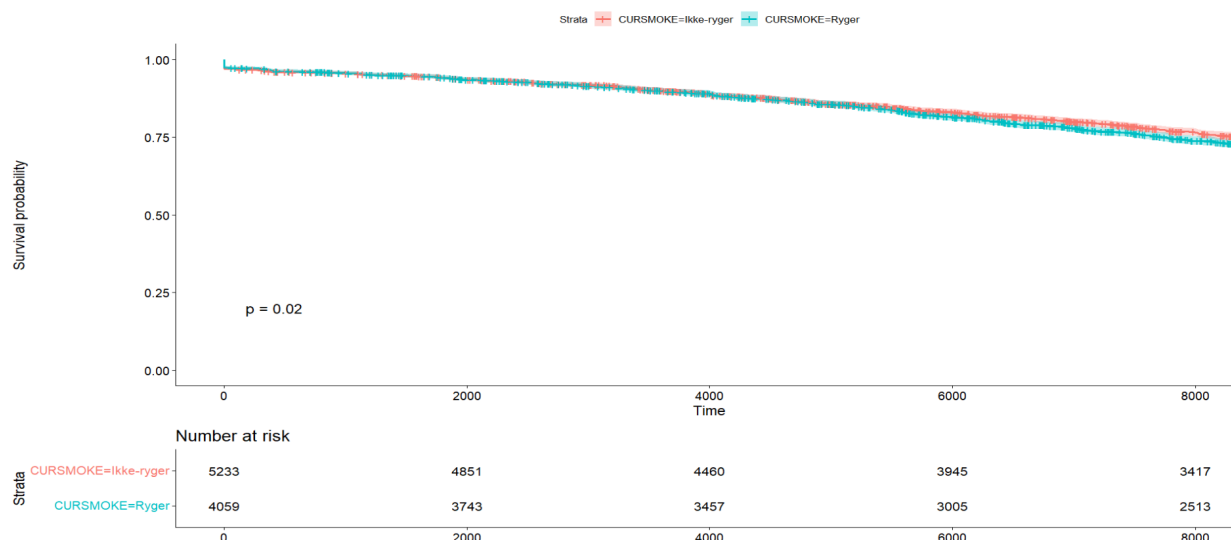


For AGE, SYSBP, TOTCHOL and BMI is close to the middle which means that there ain't a big influence on effect of getting CVD. BMI also have a confidence interval which include 1. That means it is not significant.

Diabetes is to the right but the confidence interval include 1 so its not significant.

SEX is to the left of 1, which means that it is a significant variable. SEX has a larger confidence interval but does not include 1, which means that it is significant. so it seems that women have a lower chance at getting CVD.

We can now look at specific data points and see the difference between smokers and non-smokers and between males and females. For smokers and non-smokers we get:



We can see a slower decline in survival probability among non-smokers compared to smokers. The log-rank test gives $p = 0.02$, confirming a small but statistically significant difference. **Model fit.** Visual inspection

and proportional hazards checks showed no major violations, although the slope for AGE may vary slightly over time. Overall, the model fits the data well. **Connection to previous sections.** These findings are

consistent with the logistic regression and simulation results from the earlier sections: AGE, SEX, smoking, and diabetes have the same direction of effect and similar relative strengths. However, the Cox model provides a continuous-time view of the event risk rather than a fixed-time probability. In summary, the

fitted survival model confirms that older age, male sex, higher blood pressure, smoking, and diabetes all significantly increase the risk of developing CVD, in line with both theoretical expectations and the earlier simulated findings.

6 Conclusion

We set out to understand and investigate how key clinical and demographic factors influence the development and timing of cardiovascular disease (CVD) using data from the Framingham Heart Study. Through exploratory analysis, logistic regression, and survival modelling, we examined both the probability of experiencing CVD and how that risk changes over time.

The exploratory data analysis confirmed that the dataset was well-structured, with realistic variable ranges and patterns that aligned with expectations from cardiovascular research. Logistic regression results showed that age, male sex, smoking, high blood pressure, and diabetes are major predictors of CVD, while other factors such as BMI and cholesterol contributed more modestly. Non-linear terms provided limited improvement, suggesting that the main effects were already well captured by a linear specification.

The theoretical link between logistic and survival models helped bridge our understanding of how fixed-time risk models relate to continuous-time survival analysis. The simulation results supported this connection, showing that logistic slopes remain stable across time horizons and that censoring, especially when dependent on covariates, can bias logistic estimates, whereas the Cox model remains robust.

Applying a Cox regression model to the Framingham data produced results consistent with both theory and prior analyses. The model fitted the data well, identifying the same key risk factors and achieving good predictive performance, with a concordance index around 0.72. The survival curves clearly illustrated the higher risk faced by smokers, diabetics, and older individuals, reinforcing the patterns found in the logistic models.

Overall, our findings demonstrate that logistic regression and survival models complement each other in explaining cardiovascular risk: logistic models quantify the likelihood of disease occurrence, while survival analysis captures the timing and progression of that risk. Together, they provide a coherent understanding of how lifestyle and biological factors shape cardiovascular outcomes in the Framingham population.